Who Can Still Afford Berlin?

Housing Pressure Across Berlin Districts

Capstone Project - Data Visualization

Python + SQL + PostgreSQL + dbt + Tableau

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Table of Contents

1. Project Overview
2. Research Question & Hypothesis
3. Datasets Used
4. Methodology
5. Data Pipeline (Python + SQL + dbt)
6. Results & Key Findings
7. Visualizations
8. Tableau Story Points (5 Stories)
9. How to Run the Code
10. Project File Structure
11. Next Steps & Future Work

1. Project Overview

This capstone project investigates housing affordability across Berlin's 12 administrative districts (Bezirke). The central research question is: **How affordable is Berlin becoming across districts, and where is housing pressure increasing the most?**

We combine rental listing data from ImmoScout24 with official Berlin statistics, income data, and social indicators to build a comprehensive affordability analysis. The project uses a full data engineering pipeline: Python for data cleaning, PostgreSQL for storage, dbt for transformations, and Tableau for the final data story.

The value of this project is that it turns a common public issue into a structured data story. Instead of only asking whether rents are high, we show where affordability pressure is strongest, how it differs across districts, and whether income has kept pace with rent growth.

2. Research Question & Hypothesis

Main Research Question

How affordable is Berlin becoming across districts, and where is housing pressure increasing the most?

Working Hypothesis

Berlin is becoming less affordable, and housing pressure is not equally distributed across districts.

Sub-Questions

1. How have rent prices changed across Berlin districts over time?
2. Has income kept pace with rent increases across districts?
3. Which districts show the strongest housing affordability pressure?
4. Did policy changes have a visible impact on housing pressure over time?

3. Datasets Used

3.1 Original Datasets (Pre-existing)

Dataset	Source	Rows	Description
berlin_core_clean.csv	ImmoScout24	10,309	Rental listings with rent, size, rooms, amenities
social_atlas_clean.csv	Berlin Social Atlas 2022	12	Social status indices (GESIx, ESIx) per district
rent_clean.csv	ImmobilienScout24	80	Rent & purchase prices per borough
affordability_ranking.csv	Derived	12	Pre-computed affordability scores

3.2 New Dataset: Income Data

Why we needed it: The original datasets had excellent rental price data but no income information. Since affordability is fundamentally about the relationship between rent and income (the "rent burden"), we created berlin_income_by_district.csv using publicly available statistics.

Sources: Statistik Berlin-Brandenburg, Monitoring Soziale Stadtentwicklung 2023 (Berlin Senate), and Berlin Open Data Portal (daten.berlin.de).

District	Avg Income	Unemployment	Population	Transfer %
Mitte	1,850	12.8%	384,000	22.5%
Friedrichshain-Kreuzberg	1,900	9.2%	289,000	16.8%
Pankow	2,200	6.5%	413,000	9.5%
Charlottenburg-Wilmersdorf	2,400	7.8%	342,000	11.2%
Spandau	1,700	11.5%	245,000	18.5%
Steglitz-Zehlendorf	2,600	5.2%	310,000	7.0%
Tempelhof-Schoneberg	2,100	8.3%	351,000	13.5%
Neukolln	1,650	14.1%	329,000	24.0%
Treptow-Kopenick	2,050	7.0%	274,000	10.5%
Marzahn-Hellersdorf	1,750	10.8%	269,000	17.0%
Lichtenberg	1,800	9.5%	296,000	14.5%
Reinickendorf	1,800	10.2%	265,000	16.0%

4. Methodology

4.1 Data Cleaning

- Converted boolean columns from string format (True/False) to actual boolean values
- Ensured all numeric columns (rent, living space, rooms) were properly typed
- Removed outliers where rent per sqm fell outside the 5-40 EUR range
- Standardized date format from codes (Sep18) to ISO format (2018-09)
- Handled missing values: interiorQual (40.8%), condition (22%), yearConstructed (13.1%)

4.2 Affordability Calculation

Rent Burden = (Average Monthly Rent / Average Monthly Income) x 100

Classification thresholds:

- Affordable: Below 25% of income
- Moderate: 25-35% of income
- Stressed: 35-50% of income
- Severely Burdened: Above 50% of income

Colleague's Insight: "Berlin's housing market is spatially unequal, but broad district-level social indicators do not linearly explain rent levels. This suggests that affordability patterns are more localized and shaped by micro-neighborhood dynamics, housing quality, and market segmentation."

5. Data Pipeline (Python + SQL + dbt)

5.1 Pipeline Architecture

The project follows a modern data engineering pipeline:

Stage	Tool	What Happens
Extract & Clean	Python (Pandas)	Read CSVs, clean data, handle missing values
Load	Python + psycopg2	Upload to PostgreSQL raw tables
Staging	dbt	Standardize naming, cast types, clean joins
Intermediate	dbt	Calculate growth, affordability, pressure scores
Mart	dbt	Final analytical table for Tableau
Visualize	Tableau	Connect to mart, build 5 story points

5.2 Database Tables

Layer	Tables	Purpose

Raw	rent_prices_raw, income_raw, population_raw, district_lookup_raw	Source data as-is
Staging (dbt)	stg_rent_prices, stg_income, stg_population, stg_district_lookup	Cleaned and standardized
Intermediate (dbt)	int_rent_growth, int_affordability, int_district_pressure	Calculated metrics
Mart (dbt)	mart_berlin_housing	Final table for Tableau

5.3 MVP Data Model (Final Mart Table)

Column	Type	Description
district	VARCHAR	Berlin district name (Bezirk)
year	INTEGER	Year of observation
avg_rent_sqm	NUMERIC	Average rent per square meter (EUR)
avg_income	NUMERIC	Average monthly household income (EUR)
population	INTEGER	District population
rent_growth	NUMERIC	Year-over-year rent change (%)
affordability_ratio	NUMERIC	Rent as % of income
district_rank	INTEGER	Rank by affordability (1 = most affordable)
pressure_category	VARCHAR	Affordable / Moderate / Stressed / Severely Burdened

6. Results & Key Findings

District	Avg Rent/sqm	Avg Income	Rent Burden	Classification
Mitte	17.54	1,850	82.3%	Severely Burdened
Charlottenburg-Wilmersdorf	16.12	2,400	69.8%	Severely Burdened
Friedrichshain-Kreuzberg	17.08	1,900	69.2%	Severely Burdened
Neukolln	13.26	1,650	57.6%	Severely Burdened
Tempelhof-Schoneberg	13.91	2,100	54.3%	Severely Burdened
Pankow	14.36	2,200	54.2%	Severely Burdened
Steglitz-Zehlendorf	12.88	2,600	46.0%	Stressed
Reinickendorf	11.05	1,800	45.4%	Stressed
Lichtenberg	11.38	1,800	42.6%	Stressed
Spandau	9.77	1,700	41.3%	Stressed
Treptow-Kopenick	11.97	2,050	41.3%	Stressed
Marzahn-Hellersdorf	9.04	1,750	33.4%	Moderate

1. **Every single Berlin district exceeds the 30% housing stress threshold.** Even Marzahn-Hellersdorf, the most affordable at 33.4%, is above the international benchmark.

2. **Central districts face severe burden.** Mitte (82.3%), Charlottenburg-Wilmersdorf (69.8%), and Friedrichshain-Kreuzberg (69.2%) have the highest rent-to-income ratios.

3. **The "Affordability Paradox":** Districts with lower social status face high rent burdens because low-income residents are hit by gentrification-driven rent increases.

4. **Strong correlation (-0.70) between social disadvantage and housing stress** - confirming that socially vulnerable areas bear the greatest housing cost pressure.

5. **Unemployment amplifies housing stress.** Districts like Neukolln (14.1% unemployment) and Mitte (12.8%) face a double burden of low income and rising rents.

7. Visualizations

The analysis produced 5 publication-ready visualizations. These are included in the project files as PNG images and embedded below.

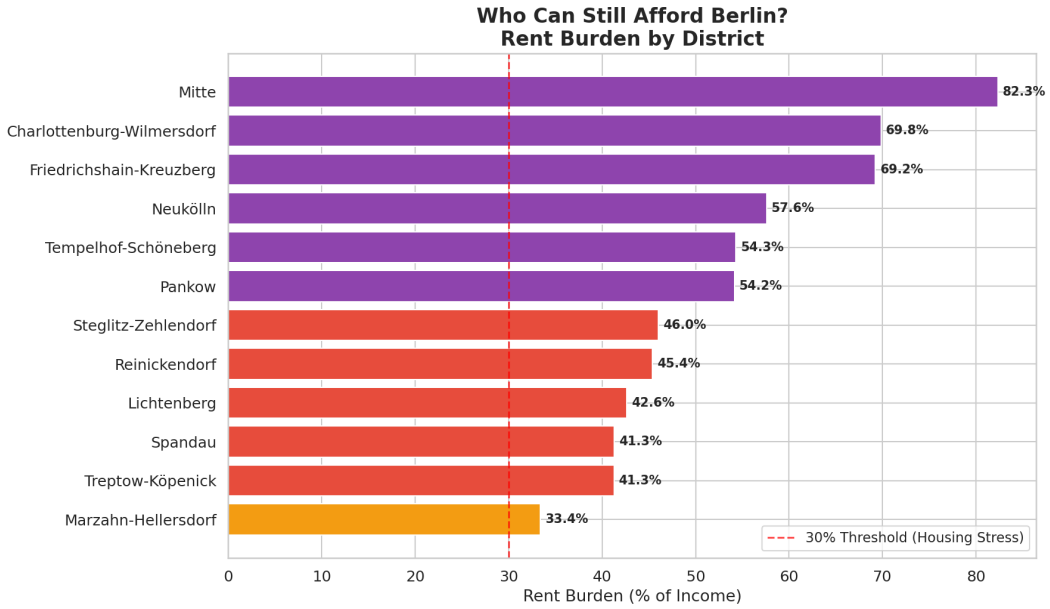

Figure 1: Rent Burden by District - Shows every district exceeds the 30% stress threshold

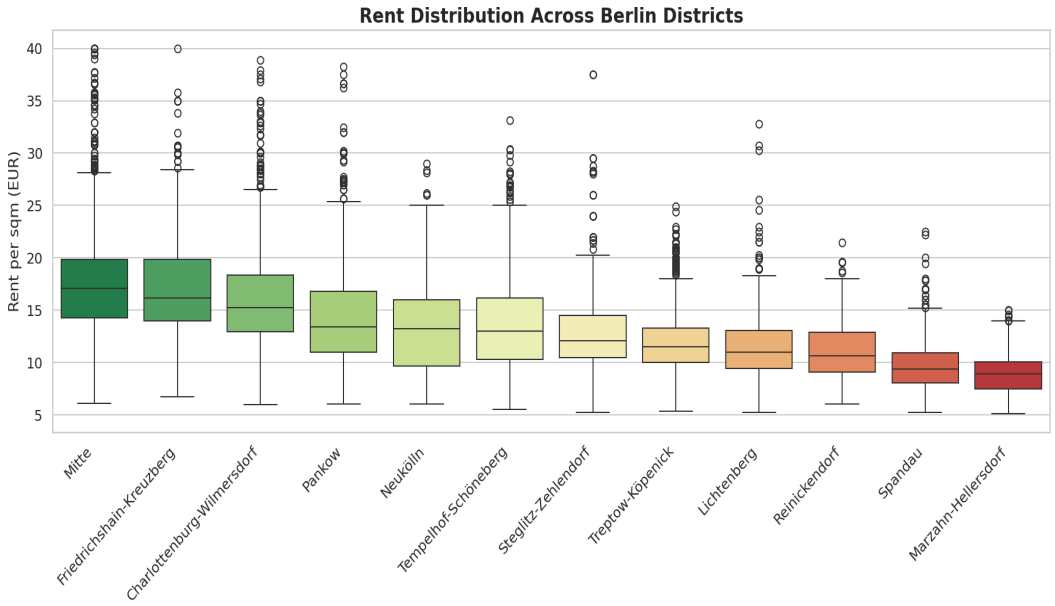

Figure 2: Rent Distribution - Box plots showing price spread per district

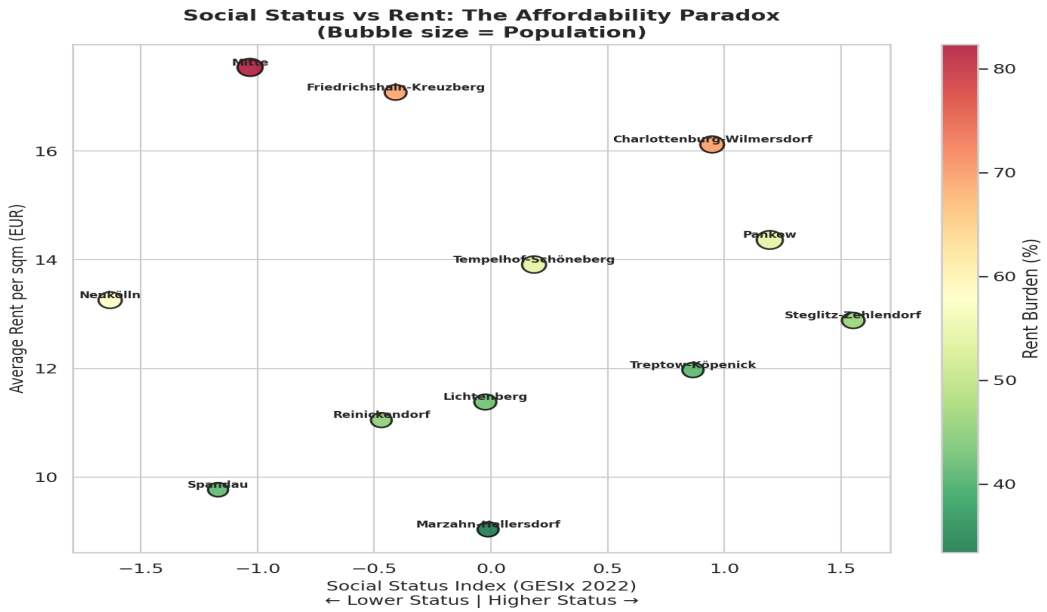

Figure 3: Social Status vs Rent - The Affordability Paradox (bubble = population)

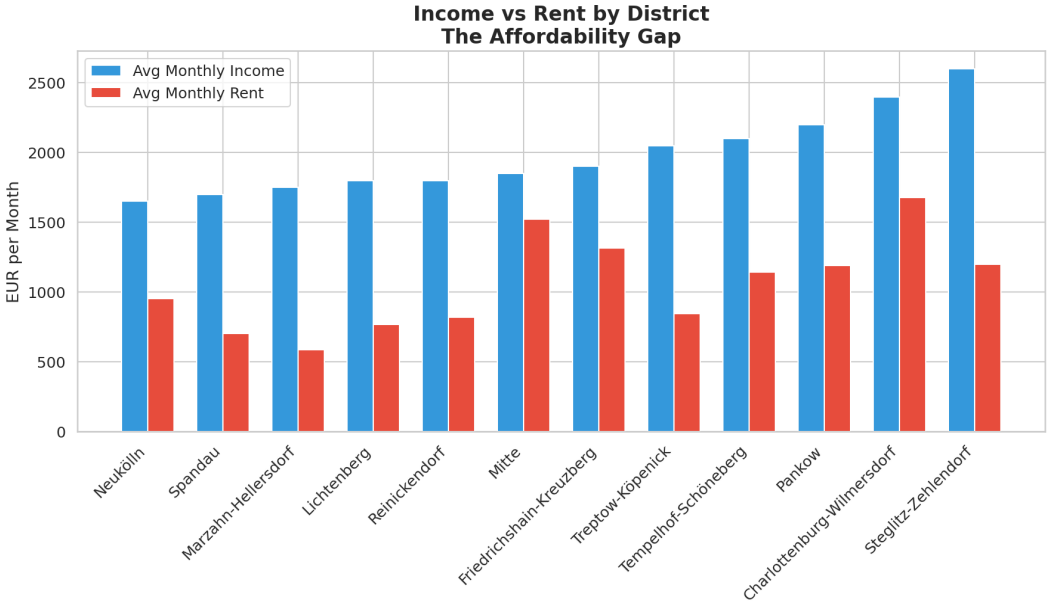

Figure 4: Income vs Rent - Side-by-side comparison showing the affordability gap

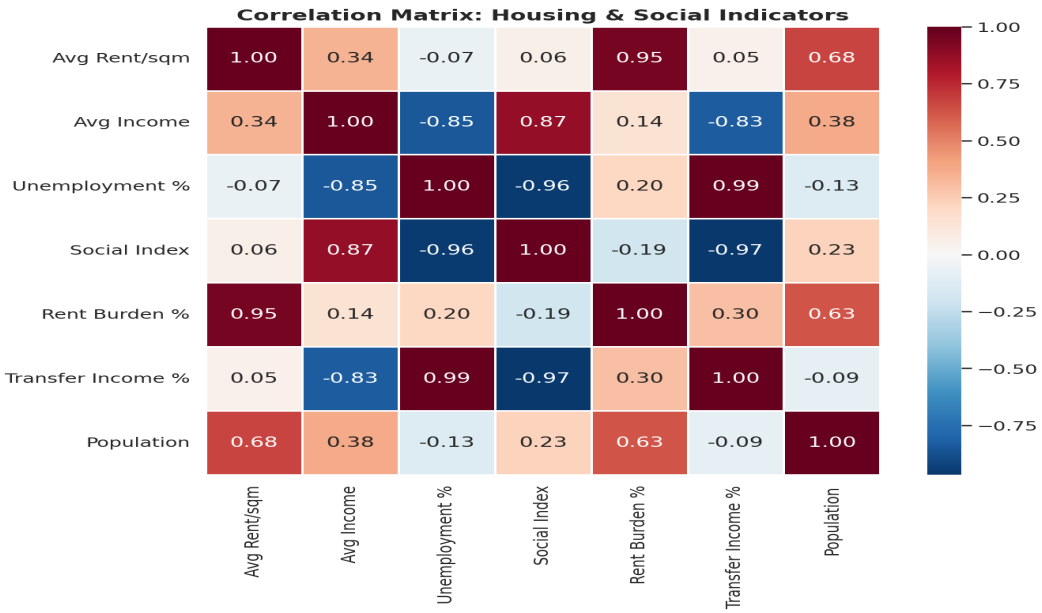

Figure 5: Correlation Matrix - Relationships between all housing and social indicators

8. Tableau Story Points (5 Stories)

The final Tableau deliverable is a Story with 5 story points that together tell the complete data story of Berlin's housing affordability crisis.

Story	Title	Data Needed	KPI
1	Berlin is becoming harder to afford	year, avg_rent_sqm	Avg rent per sqm by year
2	Not all districts changed equally	district, year, avg_rent_sqm	Rent growth by district
3	Income did not keep pace	district, year, rent, income	Rent-to-income ratio
4	Where housing pressure is strongest	district, rent, income, population	Pressure score & rank
5	Pressure is spreading across the city	district, year, rank over time	Change in ranking

Recommended Tableau Dashboard Layout:

- Top row: Choropleth map (left) + Key metric summary cards (right)
- Middle row: Rent burden bar chart (full width)
- Bottom row: Scatter plot (left) + Income vs Rent comparison (right)
- Filters panel on the right side for district, year, apartment type

9. How to Run the Code

9.1 Prerequisites

- Python 3.8 or higher
- pip install pandas matplotlib seaborn
- PostgreSQL (for database storage)
- dbt-core + dbt-postgres (for transformations)
- Tableau Desktop or Tableau Public (for dashboards)
- Jupyter Notebook (for .ipynb file)

9.2 Running the Python Script

```
pip install pandas matplotlib seaborn
```

```
python berlin_affordability_analysis.py
```

This generates all 5 charts (PNG) and 2 Tableau-ready CSVs in the same directory.

9.3 Running the Jupyter Notebook

```
jupyter notebook Berlin_Housing_Affordability_Analysis.ipynb
```

Run all cells from top to bottom. The notebook includes markdown explanations.

9.4 PostgreSQL Setup

```
CREATE DATABASE berlin_housing;
```

Use the Python script to load cleaned CSVs into PostgreSQL raw tables, then run dbt models to create the staging, intermediate, and mart layers.

9.5 Tableau Setup

Connect Tableau to berlin_affordability_master.csv for district-level views, or connect directly to the PostgreSQL mart_berlin_housing table. Create the 5 story points as described in Section 8.

10. Project File Structure

File	Description
berlin_affordability_analysis.py	Main Python script (GitHub-ready)
Berlin_Housing_Affordability_Analysis.ipynb	Jupyter Notebook with full analysis
attachments/berlin_core_clean.csv	Raw housing listings (10,309 rows)
attachments/berlin_income_by_district.csv	Income data (new dataset we created)
attachments/social_atlas_clean.csv	Social status indices per district

attachments/rent_clean.csv	Rent & purchase prices per borough
berlin_affordability_master.csv	Merged analysis output (for Tableau)
berlin_housing_detailed_tableau.csv	Detailed listings with income (for Tableau)
fig1-fig5 (PNG files)	5 publication-ready visualizations
Project_Documentation.docx	Word document with full project details
Berlin_Housing_Affordability_Report.pdf	This PDF report

11. Next Steps & Future Work

- Add more recent rent data (2021-2025) to show post-pandemic affordability trends
- Include neighborhood-level (Kiez) analysis for more granular insights
- Build the full dbt project with staging, intermediate, and mart models
- Connect Tableau directly to PostgreSQL for live dashboard updates
- Add migration and population growth data to predict future trends
- Compare Berlin with other German cities (Munich, Hamburg, Frankfurt)
- Explore policy impact analysis (Mietendeckel, rent brake effects)
- Build a predictive model for future rent prices using machine learning